

# MATH 330 Homework 5

## Functions

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**Exercise 1.** Let  $A$  be a nonempty set. Explain why there are no functions from  $A$  to  $\emptyset$ .

**Exercise 2.** Is standard division a binary operation on  $\mathbb{N}$ ? On  $\mathbb{R}$ ? Justify your answer in each case.

**Exercise 3.** Is the dot product a binary operation on  $\mathbb{R}^3$ ? Recall that the dot product is defined by  $(a, b, c) \cdot (d, e, f) = ad + be + cf$ .

**Exercise 4.** Determine whether or not each of the following binary operations on  $\mathbb{R}$  is (a) commutative, (b) associative.

1.  $a * b = |a - b|$ .

2.  $a * b = \frac{a}{b^2 + 1}$ .

3.  $a * b = \max\{a, b\}$ .

**Exercise 5.** Let  $f : X \rightarrow Y$  be a function; let  $A$  and  $B$  be subsets of  $X$ . For  $Z \subseteq X$  define  $f(Z) = \{f(z) \mid z \in Z\}$ . Determine which of the following are true. If a statement is true, prove it. If a statement is false, find a counterexample.

1.  $f(A \cup B) \subseteq f(A) \cup f(B)$

2.  $f(A \cup B) \supseteq f(A) \cup f(B)$

3.  $f(A \cap B) \subseteq f(A) \cap f(B)$

4.  $f(A \cap B) \supseteq f(A) \cap f(B)$

5.  $f(X \setminus A) \subseteq Y \setminus f(A)$

6.  $f(X \setminus A) \supseteq Y \setminus f(A)$

**Exercise 6.** For each of the false statements in the previous exercise, determine whether or not they are true under the following conditions. Prove or give a counterexample in each case.

1.  $f : X \rightarrow Y$  is an injective function.

2.  $f : X \rightarrow Y$  is a surjective function.

**Exercise 7.** Let  $A = \{1, 2, 3\}$ . For each of the following, either find a function satisfying the indicated properties or prove that no such function exists.

1. A bijective function  $f : A \rightarrow A$  other than the identity function.

2. An injective function  $g : A \rightarrow A$  that is not surjective.
3. A surjective function  $h : A \rightarrow A$  that is not injective.
4. A function  $j : A \rightarrow A$  that is neither injective nor surjective.

**Exercise 8.** Let  $A = \{1, 2, 3\}$  and  $B = \{4, 5\}$ . For each of the following, either find a function satisfying the indicated properties or prove that no such function exists.

1. A bijective function  $f : A \rightarrow B$ .
2. An injective function  $g : A \rightarrow B$  that is not surjective.
3. A surjective function  $h : A \rightarrow B$  that is not injective.
4. A function  $j : A \rightarrow B$  that is neither injective nor surjective.

**Exercise 9.** For each of the following, either find a function satisfying the indicated properties or prove that no such function exists.

1. A bijective function  $f : \mathbb{N} \rightarrow \mathbb{N}$  other than the identity.
2. An injective function  $g : \mathbb{N} \rightarrow \mathbb{N}$  that is not surjective.
3. A surjective function  $h : \mathbb{N} \rightarrow \mathbb{N}$  that is not injective.
4. A function  $j : \mathbb{N} \rightarrow \mathbb{N}$  that is neither injective nor surjective.

**Exercise 10.** Assume the Axiom of Choice. Let  $A$  be a nonempty set and  $\sim$  an equivalence relation on  $A$ ; let  $\widehat{A}$  be the set of all equivalence classes. Prove that there is an injective function  $f : \widehat{A} \rightarrow A$ .

**Exercise 11.** Find examples of sets  $X$ ,  $Y$ , and  $Z$  and functions  $f : X \rightarrow Y$  and  $g : Y \rightarrow Z$  so that  $g \circ f : X \rightarrow Z$  is surjective but  $f : X \rightarrow Y$  is not surjective.

**Exercise 12.** The standard integer addition operation  $+$  is a function from  $\mathbb{Z}^2 \rightarrow \mathbb{Z}$ . Show that if  $a \equiv_5 b$  and  $c \equiv_5 d$ , then  $a + c \equiv_5 b + d$ . This shows that addition is well-defined with respect to this equivalence relation.

**Exercise 13.** Let  $f : X \rightarrow Y$  be an invertible function and let  $f^{-1} : Y \rightarrow X$  denote the inverse of  $f$ . Show that  $f^{-1}$  is bijective and that  $f$  is the inverse of  $f^{-1}$ .

**Exercise 14.** Let  $f : X \rightarrow Y$  be a function and suppose that the function  $g : Y \rightarrow X$  is the inverse of  $f$ . Show that the composition  $g \circ f : X \rightarrow X$  is the identity function, i.e. that  $g(f(x)) = x$  for every  $x \in X$ .